A320 FAL TWIN · DECISION TRACE

FINAL ASSEMBLY LINE S50 → FLIGHT LINE · INTERACTIVE CONCEPT

One-pager · October 2026 · Miguel Figueroa
github.io · a320-fuselage-agent-control/fal-twin

SYNTHETIC DATA

GOAL

WHY THIS TWIN EXISTS

Achieve a production rate of **100 aircraft per month** by optimizing end-to-end manufacturing operations, identifying the **critical path**, eliminating **non-value-added activities**, and implementing **Lean Manufacturing** principles, including Kanban, Just-in-Time (JIT), and Just-in-Sequence (JIS), across all assembly stages.

Leverage Edge AI, Digital Twins, Industrial Metaverse, and Cloud technologies to enable real-time monitoring, predictive analysis, and early detection of supply chain, quality, and production disruptions. Provide stakeholders with actionable insights and timely alerts to accelerate decision-making, minimize downtime, reduce scrap and rework, and improve overall production efficiency.

100/mo is the programme rate across all lines; the modelled line's share at takt 2.0 d is 10.5/mo (target below).

KEY RESULTS · NOW VS TARGET

TYPICAL RUN · TARGETS ILLUSTRATIVE

KPI	NOW	TARGET	LEVER IN THE TWIN
Delivery rate	9.8 /mo	≥ 10.5	third S20 shift, floating team
Lead time	23.6 d	≤ 26.7 d	critical path, line rebalancing
On-time delivery	100 %	≥ 90 %	late-MSN alerts, rush-order planner
First-pass yield	38 %	≥ 80 %	edge-AI hole inspection, kit gate
Rework	19 h/ac	≤ 8 h	edge-AI inspection at S41
Gliders	0	0	engine-supply alert, expedite
OEE line	33 %	≥ 60 %	quality levers lift Q, balance lifts A
Power peak · 24 h	2,630 kW	≤ 2,650	stagger cure ovens, battery shaving

INDUSTRIAL METAVERSE

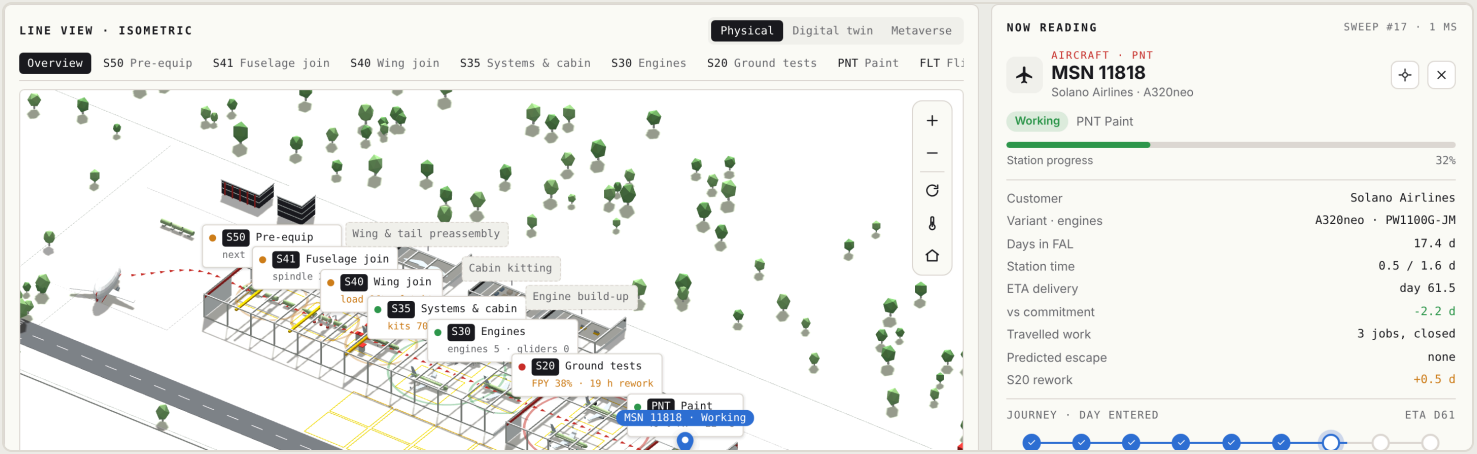
Enable immersive 3D visualization and cross-functional collaboration to evaluate production processes, test improvements, and accelerate decision-making.

DIGITAL TWIN VIEW

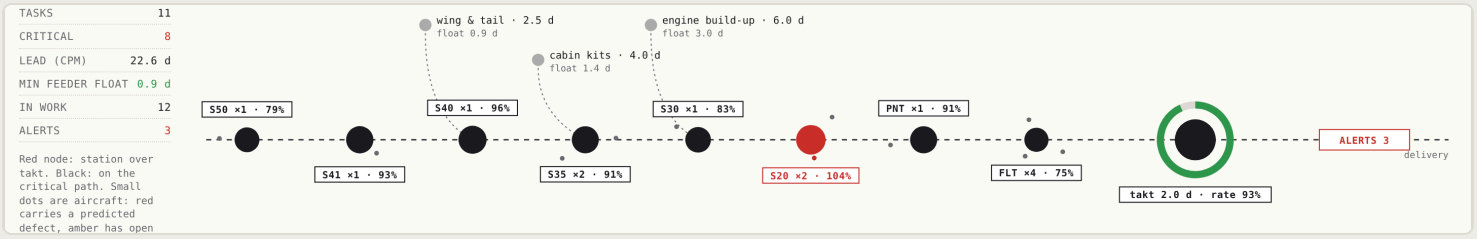
Simulate production scenarios to anticipate disruptions, eliminate waiting times, and optimize throughput. Evaluate the impact of proposed changes on lead time, quality, and resource utilization before physical implementation.

EDGE AI

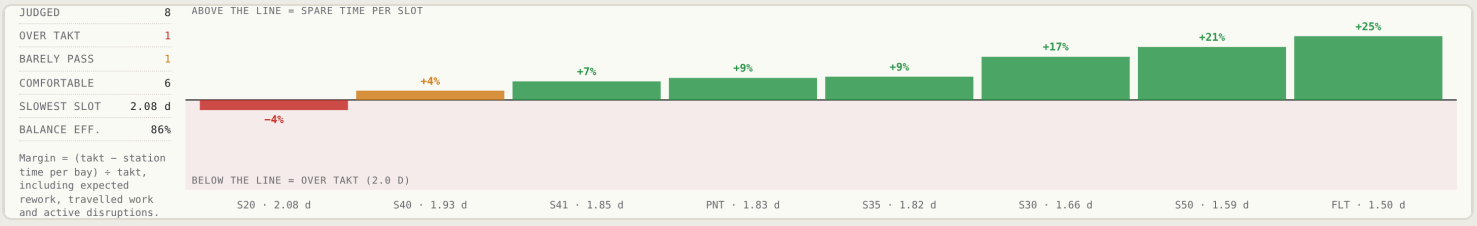
Leverage real-time sensor data and on-site AI processing to detect anomalies, identify bottlenecks, prevent defects, and reduce scrap, improving Right First Time (RFT) performance and overall production efficiency.



Line view (isometric 3D, physical · digital twin · metaverse · heat) beside "Now reading": the selected aircraft with status, ETA vs commitment and its station-by-station journey.



Relation graph: feeders → stations → takt → alerts. Node size = load vs takt; red = over takt, black = critical path; small dots are aircraft.



Takt margin field: each station's distance to the takt line, sorted worst to best; below the line = over takt.

BUILD & RUN · one self-contained HTML page (three.js, in-browser simulation and CPM, no backend) · rule-based assistant, not a language model · keeps the screen awake for wall displays · live: **mfmadinaveitia1-del.github.io/a320-fuselage-agent-control/fal-twin/**

All layouts, cycle times, rates, defect rates, costs, customers and power figures are synthetic, invented for this concept. They do not represent any manufacturer's data. KPI values are from one typical simulation run.